

IBM Bob Java Modernization - Migrating Legacy Struts Application into a Modern React & Spring Boot Microservices

AI-Powered Application Modernization from Struts, JSP, Hibernate, and Java 6 to Spring boot Microservices and react frontend

Overview :
This solution showcases how IBM Bob analyzes, documents, and modernizes a 20-year-old Java monolithic application. Using natural-language prompts, Bob converts a legacy Struts/JSP application into a modern cloud-native solution consisting of Spring Boot microservices, a React frontend, PostgreSQL database, automated testing, DevSecOps assets, and comprehensive modernization documentation.

Business Objective :

- Accelerate legacy application modernization.
- Reduce manual reverse-engineering and migration effort.
- Modernize monolithic applications into scalable microservices.
- Improve maintainability, security, and cloud readiness.
- Generate migration artifacts, documentation, and deployment assets automatically.

Key Capabilities Demonstrated:

- Legacy application analysis and reverse engineering.
- Automated architecture and application inventory documentation.
- Domain-Driven Design (DDD) based microservice identification.
- Monolith-to-microservices transformation.
- Struts/JSP to React frontend modernization.
- PostgreSQL database migration planning and implementation.
- Automated generation of unit and integration tests.
- DevSecOps pipeline, Docker, Jenkins, and OpenShift asset generation.
- Automated creation of traceability, validation, and modernization reports.
- AI-assisted remediation of migration issues such as CORS, Flyway, and SPA routing.

Outcome:

- 3 Spring Boot 3.x Microservices (User, Vehicle, Premium), React 18 Single-Page Application (SPA) and PostgreSQL Database Migration
- Automated Test Suites and Test Documentation, 15 Comprehensive Modernization Documents, OpenShift Deployment Manifests and CI/CD Pipeline Assets
- End-to-End Modernization Delivered Through Simple AI Prompts

How It Works :

Analyze : Bob scans the legacy Struts application, identifies business logic, dependencies, data models, and technical debt.

Document: Generates architecture, SRE, DDD, traceability, migration, validation, and operational documentation.

Design: Identifies bounded contexts and proposes a target microservices architecture.

Transform: Converts legacy components into Spring Boot microservices and React frontend components.

Validate: Generates unit tests, integration tests, and migration traceability artifacts.

Deploy: Produces Docker, Jenkins, and OpenShift deployment assets ready for cloud deployment.

Refine: Iteratively fixes migration issues and enhances the solution through additional natural-language prompts.



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